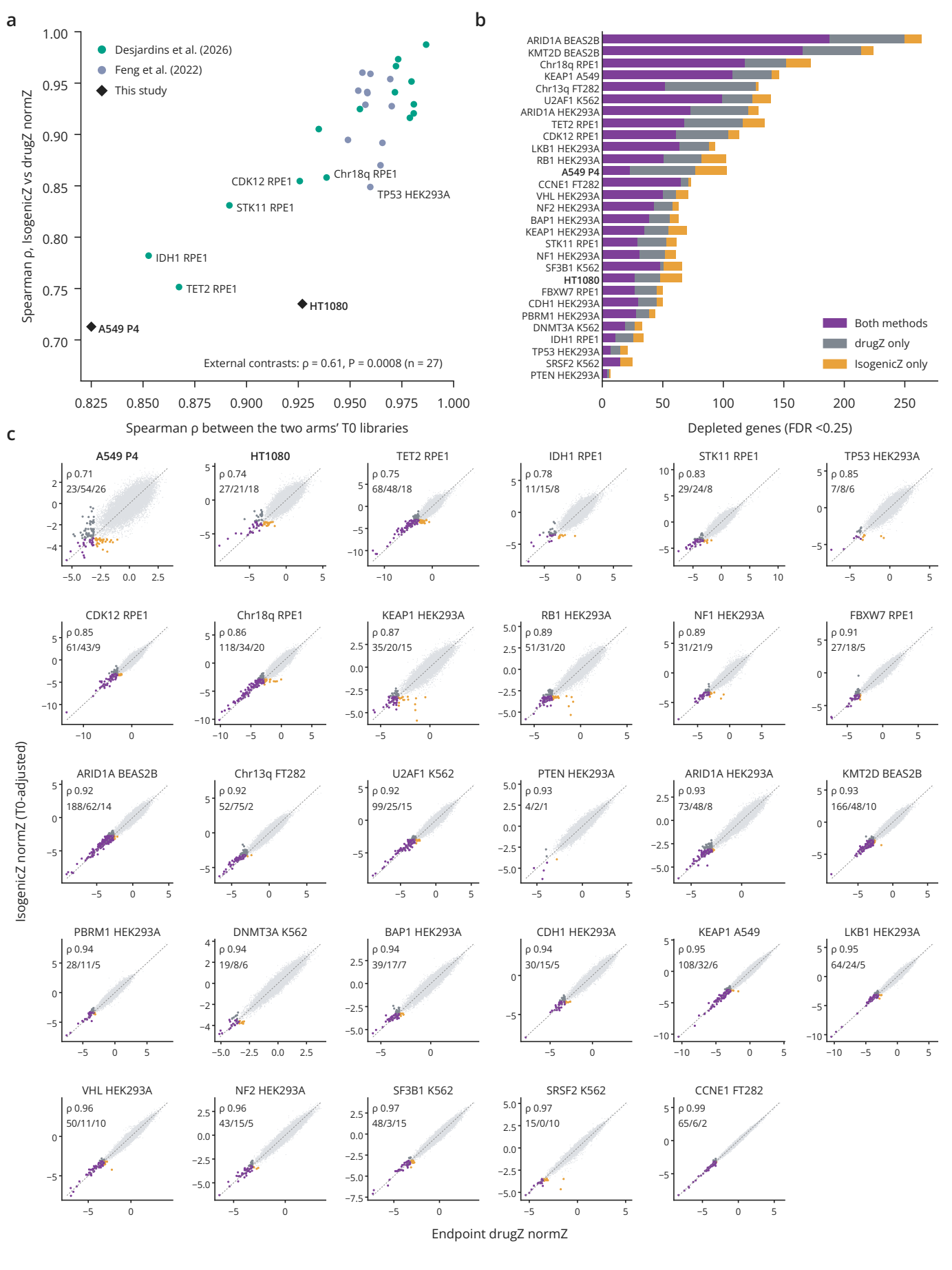

a
Desjardins et al. (2026)
Feng et al. (2022)
This study
CDK12 RPE1
Chr18q RPE1
TP53 HEK293A
STK11 RPE1
IDH1 RPE1
TET2 RPE1
HT1080
A549 P4
External contrasts: ρ = 0.61, P = 0.0008 (n = 27)
Spearman ρ, IsogenicZ vs drugZ normZ
Spearman ρ between the two arms' T0 libraries
b
ARID1A BEAS2B
KMT2D BEAS2B
Chr18q RPE1
KEAP1 A549
Chr13q FT282
U2AF1 K562
ARID1A HEK293A
TET2 RPE1
CDK12 RPE1
LKB1 HEK293A
RB1 HEK293A
A549 P4
CCNE1 FT282
VHL HEK293A
NF2 HEK293A
BAP1 HEK293A
KEAP1 HEK293A
STK11 RPE1
NF1 HEK293A
SF3B1 K562
HT1080
FBXW7 RPE1
CDH1 HEK293A
PBRM1 HEK293A
DNMT3A K562
IDH1 RPE1
TP53 HEK293A
SRSF2 K562
PTEN HEK293A
Both methods
drugZ only
IsogenicZ only
Depleted genes (FDR <0.25)
c
A549 P4 ρ 0.71 23/54/26
HT1080 ρ 0.74 27/21/18
TET2 RPE1 ρ 0.75 68/48/18
IDH1 RPE1 ρ 0.78 11/15/8
STK11 RPE1 ρ 0.83 29/24/8
TP53 HEK293A ρ 0.85 7/8/6
CDK12 RPE1 ρ 0.85 61/43/9
Chr18q RPE1 ρ 0.86 118/34/20
KEAP1 HEK293A ρ 0.87 35/20/15
RB1 HEK293A ρ 0.89 51/31/20
NF1 HEK293A ρ 0.89 31/21/9
FBXW7 RPE1 ρ 0.91 27/18/5
ARID1A BEAS2B ρ 0.92 188/62/14
Chr13q FT282 ρ 0.92 52/75/2
U2AF1 K562 ρ 0.92 99/25/15
PTEN HEK293A ρ 0.93 4/2/1
ARID1A HEK293A ρ 0.93 73/48/8
KMT2D BEAS2B ρ 0.93 166/48/10
PBRM1 HEK293A ρ 0.94 28/11/5
DNMT3A K562 ρ 0.94 19/8/6
BAP1 HEK293A ρ 0.94 39/17/7
CDH1 HEK293A ρ 0.94 30/15/5
KEAP1 A549 ρ 0.95 108/32/6
LKB1 HEK293A ρ 0.95 64/24/5
VHL HEK293A ρ 0.96 50/11/10
NF2 HEK293A ρ 0.96 43/15/5
SF3B1 K562 ρ 0.97 48/3/15
SRSF2 K562 ρ 0.97 15/0/10
CCNE1 FT282 ρ 0.99 65/6/2
IsogenicZ normZ (T0-adjusted)
Endpoint drugZ normZ